

Experiment

39

Chemical properties of acids and bases

Background

Acids and bases are two classes of compounds that have wide application in industrial and household chemistry.

Acids are substances that donate protons and in aqueous solutions produce hydrogen ions (H^+). Commonly encountered acids include spirits of salts (hydrochloric acid), battery acid (sulfuric acid) and vinegar (ethanoic or acetic acid).

Bases are substances that accept protons, and in aqueous solutions produce hydroxide ions (OH^-). Soda ash (sodium carbonate), caustic soda (sodium hydroxide), washing soda (sodium carbonate-10-water) and ammonia are commonly used bases.

In this experiment you will investigate some of the important chemical properties of acids and bases.

Equipment required

Bunsen burner

Dropper

Test tubes (six)

Taper and matches

Hydrochloric acid $[HCl]$ 2 mol L^{-1} (30 mL)

Sodium hydroxide solution $[NaOH]$ 2 mol L^{-1} (30 mL)

Limewater $[Ca(OH)_2]$ (20 mL)

Vinegar [3–4% CH_3COOH] (2 mL)

Two pea-sized pieces of aluminium $[Al]$, copper $[Cu]$, iron $[Fe]$, magnesium $[Mg]$ and zinc $[Zn]$

About 1 g each of:

Copper(II) oxide $[CuO]$

Sodium hydrogencarbonate $[NaHCO_3]$

Ammonium chloride $[NH_4Cl]$

Calcium carbonate (marble chips) $[CaCO_3]$

Sodium ethanoate (sodium acetate) $[NaCH_3COO]$

Sodium sulfide $[Na_2S]$

Sodium sulfite $[Na_2SO_3]$

Blue litmus paper

Red litmus paper

Stopper fitted with gas delivery tube

Procedure

#1 Draw up a table which can be used to summarise the experiments you will perform. Your table should include the following headings:

- a Type of reaction
- b The chemicals involved in the reaction you are performing
- c Your observations of the reaction
- d An explanation of your observations which should, where possible, include an equation.

A Effect of Acids and Bases on Litmus

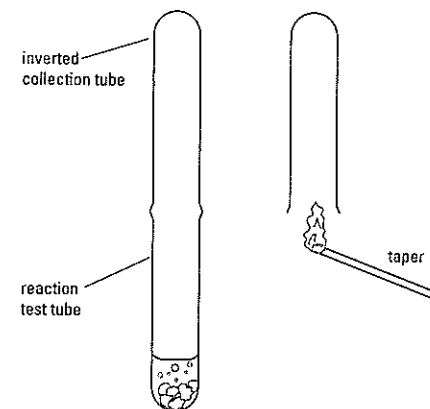
#1 Place a drop of 2 mol L^{-1} HCl onto pieces of red and blue litmus paper. Repeat using 2 mol L^{-1} $NaOH$.

B Reaction of Acids with Metals

#1 Place pea-sized pieces of Mg , Zn , Al and Cu into separate test tubes. Add 2–3 mL of 2 mol L^{-1} HCl to each. If there is no sign of reaction warm the mixture carefully.

#2 If a gas is evolved, collect it by inverting another test tube over the mouth of the reaction tube. Keep the collection tube inverted and hold a lighted taper near the open end as shown in Figure 39.1.

Fig. 39.1



C Reaction of Bases With Metals

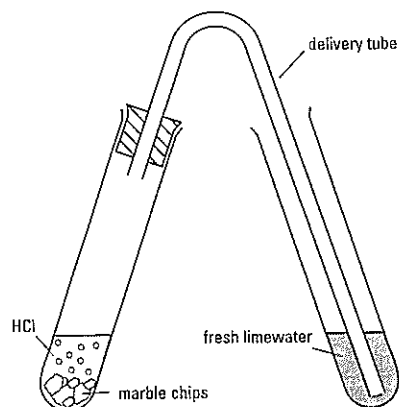
#1 Repeat the procedure in part B but use 2 mol L^{-1} $NaOH$ instead of HCl .

D Reaction of Acids with Metal Oxides

#1 Place a pea-sized quantity of CuO in a test tube. Add 2–3 mL of 2 mol L^{-1} HCl , heat the mixture and allow it to stand.

#2 Observe the colour of the solution produced.

Fig. 39.2



E Reaction of Strong Acids with Anions of Weak Acids

1 Carbonates

- #1 Add 2–3 mL of 2 mol L⁻¹ HCl to some marble chips in a test tube.
 #2 Pass some of the gas produced into limewater as shown in Figure 39.2.

2 Hydrogencarbonates

- #3 Repeat the procedure used above but replace the marble chips with NaHCO₃.

3 Sulfides

Safety Note

- The hydrogen sulfide gas produced by this reaction is extremely poisonous and should not be breathed in.
- Conduct this part of the experiment in a fumehood.

- #4 Add 2–3 mL of 2 mol L⁻¹ HCl to a small quantity of solid Na₂S. The gas produced has a characteristic odour often likened to that of rotten eggs.

4 Sulfites

Safety Note

- Sulfur dioxide is poisonous and should not be breathed in large quantities.
- Use your hand to waft a small amount of gas towards your nose.

- #5 Add 2–3 mL of 2 mol L⁻¹ HCl to a small quantity of sodium sulfite. Warm gently.
 #6 Test the gas evolved using moist red and blue litmus paper.
 #7 Carefully sniff the gas by wafting some of it towards your nose.

5 Ethanoates (acetates)

- #8 Add 2–3 mL of 2 mol L⁻¹ HCl to a small quantity of solid sodium ethanoate (sodium acetate), and heat the mixture.
 #9 Smell the resulting solution and identify the substance produced. Compare the smell with that of vinegar.

F Neutralisation of a Strong Acid with a Strong Base

- #1 Place 10 drops of 2 mol L⁻¹ HCl into a test tube and add a small piece of blue litmus paper. Note the colour of the litmus paper.
 #2 Add 2 mol L⁻¹ NaOH dropwise until the litmus paper changes colour.

G Reaction of Strong Bases with Cations of Weak Bases

Safety Note

- Ammonia has an irritating odour and should not be breathed in large quantities.
- Use your hand to waft a small amount of gas towards your nose.

- #1 Add 2–3 mL of 2 mol L⁻¹ NaOH to a small quantity of solid ammonium chloride and warm the mixture gently until it is almost boiling.
 #2 Test the gas evolved using moist red and blue litmus paper.
 #3 Carefully sniff the gas by wafting some of it towards your nose.

Processing of results, and questions

- 1 Summarise the properties you have found about dilute acids and bases by completing the following general equations:

Acid + Reactive Metal	→
Acid + Metal Oxide	→
Acid + Carbonate	→
Acid + Hydrogencarbonate	→
Acid + Sulfide	→
Acid + Sulfite	→
Acid + Ethanoate (Acetate)	→
Acid + Strong Base	→
Base + Amphoteric Metal	→
Base + Ammonium Salt	→